Tommy Kuan

Engineering Manager

Profile

Engineering manager scaling teams and systems the AI-native way: spec-driven delivery with coding agents, real-time features for 5,000+ concurrent players, and an auth platform behind every Sezzle login, built hands-on before I moved into management.

Experience

Engineering Manager, SHOPX at Sezzle, Minneapolis, MN

Apr 2025 — Present

- Lead a 13-engineer team owning Sezzle's rewards, community, and gamification products across Go microservices and React Native.
- Launched the Rewards tab and Sezzle Points; shoppers earn through cashback offers, play-to-earn games, surveys, and financial-education content. Earning shoppers return 35% more often week over week.
- Built ShopX Gameshow, a live-streamed trivia experience (AWS IVS, SSE, React Native), peaking at 5,000+ concurrent players across 6 shows with prize pools up to $4,000.
- Moved the team to AI-agent-driven development with spec-first blueprints as shared context for building and review; streamlined SLO alert triage with LLM agents.

Software Engineer at Sezzle, Minneapolis, MN

Jun 2021 — Apr 2025

- Built AuthV5, Sezzle's shopper authentication platform, end to end: a Go API for phone, OTP, PIN, and signup flows plus a shared React auth package. Rolled it out across Checkout, the customer dashboard, and Sezzle Card, then retired the legacy system.
- Built passkey (WebAuthn) biometric login end to end, from the Go API and credential storage to the dashboard registration and login flows.
- Owned shopper account security: suspicious-login detection and alerts, IP rate limiting, log-out-all-devices, change PIN, self-service phone change, and duplicate-account prevention.

Research & Teaching

Research Assistant, Gravitational Wave Group at University of Minnesota

2019 — 2021

Extended the BLIP Bayesian pipeline to recover primordial gravitational-wave parameters for LISA; automated RF measurement for CMB detectors; physics TA.

Education

B.S. Physics (Computational), University of Minnesota Twin Cities

2018 — 2021

- Minor: Computer Science

Details

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How I Work

- AI-native engineering
- Spec-driven development
- Agent review guardrails
- Claude skills & MCP
- Scaling engineering teams
- Real-time systems at scale
- Auth & account security
- SQL & Redshift analytics
- A/B testing
- Product & game design

Stack

Go · TypeScript · React Native · Kubernetes · AWS · Redshift